

Configure the Internet-Facing Application Load Balancer

1. Create the Web Target Group

- Go to **Target Groups** → **Create target group**.
- Choose **Instances** as the target type.
- Name it (e.g., **MYPROJECTTG1-web-80**).
- Select **HTTP** protocol and Port **80** (Nginx routes your Web tier).
- Select your project **VPC**.
- **Configure Health Checks:**
 - Protocol: **HTTP**
 - Path: Enter your frontend homepage path (e.g., **/**).

MYPROJECTTG1 Actions ▾

Details
 `arn:aws:elasticloadbalancing:us-east-1:041944951715:targetgroup/MYPROJECTTG1/2d55a9af7dc2ef6c`

Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1	VPC vpc-081c241c4e4b95465 
IP address type IPv4	Load balancer MY-PROJECT-LB-INTERNET-FACING 		

2
Total targets

 2
Healthy
0 Anomalous

 0
Unhealthy

 0
Unused

 0
Initial

 0
Draining

► Distribution of targets by Availability Zone (AZ)
Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets

Monitoring

Health checks

Attributes

Tags

Health check settings

Protocol
HTTP

Unhealthy threshold
2 consecutive health check failures

Path
/

Timeout
5 seconds

Port
Traffic port

Interval
30 seconds

Healthy threshold
5 consecutive health check successes

Success codes
200

Edit

2. Register Web EC2 Instances

- Select your **Public (or Private Web) EC2 instances**.
- Enter Port **80** and click **Include as pending below**.
- Click **Create target group**.
- Both instance should be healthy

Targets
Monitoring
Health checks
Attributes
Tags

Registered targets (2) Info
Anomaly mitigation: Not applicable
Deregister
Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets
1

	Instance ID	Name	Port	Zone	Health status	Health status details
☐	i-018ab64307567f408	1A_WEB_SERV...	80	us-east-1a (us...	Healthy	-
☐	i-06cc180821e0d7e16	1B_WEB_SERV...	80	us-east-1b (us...	Healthy	-

3. Create the Internet-Facing ALB

- Go to Load Balancers → Create load balancer → Application Load Balancer.
- Basic Configuration:
 - Name: e.g., **MY-PROJECT-LB-INTERNET-FACING**
 - Scheme: **Internet-facing** (This provisions public IP addresses).
 - IP address type: **IPv4**

MY-PROJECT-LB-INTERNET-FACING
Actions

Details

Load balancer type Application	Status Active	VPC vpc-081c241c4e4b95465	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone Z35SXDOTRQ7X7K	Availability Zones subnet-06d67f47275c42f33 us-east-1a (use1-az1) subnet-04b908b82f753e561 us-east-1b (use1-az2)	Date created September 22, 2026, 15:57 (UTC+05:30)

Load balancer ARN
arn:aws:elasticloadbalancing:us-east-1:041944951715:loadbalancer/ap
p/MY-PROJECT-LB-INTERNET-FACING/cbf3809981d929fc

DNS name Info
MY-PROJECT-LB-INTERNET-FACING-880241944.us-east-1.elb.amazonaw
s.com (A Record)

Need to configure Listener port

Listeners and rules
Network mapping
Resource map
Security
Monitoring
Integrations
Attributes

Listeners and rules (1) Info
Manage rules
Manage listener
Add listener

A listener checks for connection requests on its configured protocol and port. Incoming requests are evaluated against rules in priority order; the first matching rule determines routing. If no rule matches, the default action is applied.

Filter listeners
1

	Protocol:Port	Default action	Rules	ARN	Security policy
☐	HTTP:80	Forward to target group MYPROJECTTG1: 1 (100%) Target group stickiness: Off	1 rule	ARN	Not applicable

For verify

Go to [http:// MY-PROJECT-LB-INTERNET-FACING-880241944.us-east-1.elb.amazonaws.com](http://MY-PROJECT-LB-INTERNET-FACING-880241944.us-east-1.elb.amazonaws.com)

Feedback Form-1

Submit

Feedback Form-2

Submit

#Conclusion:-Public Load Balancer Works properly

Configure Internal Application Load Balancer

1. Create the App Target Group

- Go to the **EC2 Console** → **Target Groups** → **Create target group**.
- Choose **Instances** as the target type.
- Name it (e.g., **MYPROJECTINTERNALTG**).
- Select **HTTP** protocol and Port **5000** (where your Flask app is running).
- Select your project **VPC**.
- **Configure Health Checks:**
 - Protocol: **HTTP**
 - Path: Enter your health check route (e.g., **/**).

MYPROJECTINTERNALTG

[Actions](#)

Details

arn:aws:elasticloadbalancing:us-east-1:041944951715:targetgroup/MYPROJECTINTERNALTG/deb343da3e99494e

Target type

Instance

Protocol : Port

HTTP: 80

Protocol version

HTTP1

VPC

[vpc-081c241c4e4b95465](#)

IP address type

IPv4

Load balancer

[INTERNALBPRIVET](#)

2

Total targets

2

Healthy

0 Anomalous

0

Unhealthy

0

Unused

0

Initial

0

Draining

► Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

2. Register Flask EC2 Instances

- Click **Next** to move to the **Register targets** screen.
- Select your **Private Application EC2 instances** from the list.
- Enter Port **5000** and click **Include as pending below**.
- Review and click **Create target group**.

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (2) [Info](#)

[Anomaly mitigation: Not applicable](#)



Deregister

Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

< 1 >

☐	Instance ID	Name	Port	Zone	Health status	Health status details
☐	i-0b5571a4fea18d82e	2B_APPLICATION...	5000	us-east-1b (us...)	Healthy	-
☐	i-02a70a790d7293e1e	2A_APPLICATION...	5000	us-east-1a (us...)	Healthy	-

Targets	Monitoring	Health checks	Attributes	Tags								
Health check settings Edit <table border="1"> <tr> <td>Protocol HTTP</td> <td>Path /</td> <td>Port Traffic port</td> <td>Healthy threshold 5 consecutive health check successes</td> </tr> <tr> <td>Unhealthy threshold 2 consecutive health check failures</td> <td>Timeout 5 seconds</td> <td>Interval 30 seconds</td> <td>Success codes 200</td> </tr> </table>					Protocol HTTP	Path /	Port Traffic port	Healthy threshold 5 consecutive health check successes	Unhealthy threshold 2 consecutive health check failures	Timeout 5 seconds	Interval 30 seconds	Success codes 200
Protocol HTTP	Path /	Port Traffic port	Healthy threshold 5 consecutive health check successes									
Unhealthy threshold 2 consecutive health check failures	Timeout 5 seconds	Interval 30 seconds	Success codes 200									

3. Create the Internal ALB

- Navigate to **Load Balancers** → **Create load balancer** → Select **Application Load Balancer**.
- **Basic Configuration:**
 - Name: e.g., alb-internal-app
 - Scheme: **Internal** (*Crucial: This keeps it private within your VPC*).
 - IP address type: **IPv4**

INTERNALLBPRIVET Actions

▼ Details

Load balancer type Application	Status Active	VPC vpc-081c241c4e4b95465	Load balancer IP address type IPv4
Scheme Internal	Hosted zone Z35SXDOTRQ7X7K	Availability Zones subnet-0330be35645c5a874 us-east-1a (use1-az1) subnet-0acf269ff6be0aa05 us-east-1b (use1-az2)	Date created September 22, 2026, 16:32 (UTC+05:30)
Load balancer ARN arn:aws:elasticloadbalancing:us-east-1:041944951715:loadbalancer/app/INTERNALLBPRIVET/761933073412e839		DNS name Info internal-INTERNALLBPRIVET-1761774612.us-east-1.elb.amazonaws.com (A Record)	

Also verify listener port

<	Listeners and rules	Network mapping	Resource map	Security	Monitoring	Integrations	Attributes	>
---	---------------------	-----------------	--------------	----------	------------	--------------	------------	---

Listeners and rules (1) Info

Manage rules Manage listener Add listener

A listener checks for connection requests on its configured protocol and port. Incoming requests are evaluated against rules in priority order; the first matching rule determines routing. If no rule matches, the default action is applied.

☐ Protocol:Port

☐ HTTP:5000

• Forward to target group

[MYPROJECTINTERNALTG](#): 1 (100%)
Target group stickiness: Off

1 rule

ARN

Not applicable

Finally verify the backend API via web tier instance

```
ubuntu@ip-192-168-1-149:~$ curl http://internal-INTERNALBPRIKET-1761774612.us-east-1.elb.amazonaws.com:5000/  
{"status":"Flask is running"}  
ubuntu@ip-192-168-1-149:~$
```